

A Six-Exponentials Obstruction to All-Prime Pressure Labels in Hénon Dynamics

Liang Wang^{*1}

¹School of Artificial Intelligence and Automation, Huazhong University of Science and Technology
Wuhan 430074, P.R. China

^{*}Corresponding author

Preprint, August 2026

Abstract

Pressure normalization turns the instability roof of a certified area-preserving Hénon survivor into an entropy-one suspension with the prime-orbit law $\Pi(T) \sim e^T/T$. Its repetition-compatible scalar labels are $P_\gamma = |\Lambda_\gamma|^{h_*}$, but their arithmetic type was open. We exhibit primitive survivor orbits of periods one, three and four whose positive unstable moduli L_1, L_3, L_4 generate a degree-32 compositum. The independent inversion automorphisms prove that $\log L_1, \log L_3, \log L_4$ are linearly independent over $\mathbb{Q}$. For rational $h > 0$, algebraic-unit integrality prevents L_j^h from being a rational prime. For irrational h , the Six Exponentials Theorem forces at least one of L_1^h, L_3^h, L_4^h to be transcendental. Thus, for every real $h > 0$, the three labels cannot all be rational primes. This unconditionally rejects an all-orbit termwise prime interpretation of the pressure labels, including the actual pressure root, while leaving the dynamical prime-orbit counting theorem intact. An exact symbolic certificate reconstructs the orbits, monodromy polynomials, ramification ladder and degree-32 primitive element.

1 The remaining scalar gate

For

$$H_6(q, p) = (1 - 6q^2 - p, q),$$

the certified four-rectangle survivor is conjugate to a mixing finite-type shift. Its positive non-lattice instability roof has a unique pressure root h_* with

$$0.277980 < h_* < 0.277987.$$

The pressure-normalized suspension has entropy one and obeys the dynamical prime-orbit theorem

$$\Pi(T) \sim \frac{e^T}{T}.$$

Its intrinsic scalar label is

$$P_\gamma = |\Lambda_\gamma|^{h_*}, \tag{1}$$

which preserves every orbit repetition. Earlier all-period algebra proves that every raw multiplier is an algebraic unit, and the rational functorial classification leaves real powers as the precise scalar survivor. The question addressed here is stronger than counting: can (1) be a rational prime for every primitive orbit?

Our result uses no conjecture about the arithmetic nature of h_* . It also does not assign primes to orbits or inspect a prime table.

2 Three exact primitive survivor orbits

Periodic coordinates obey

$$q_{i+1} = 1 - 6q_i^2 - q_{i-1}, \quad (2)$$

and the chronological derivative step is

$$J(q_i) = \begin{pmatrix} -12q_i & -1 \\ 1 & 0 \end{pmatrix}.$$

Write $X_- = [-5/8, -1/3]$ and $X_+ = [1/3, 5/8]$. In state order $(--, -, +, ++)$, with state $(\text{sgn } q_i, \text{sgn } q_{i-1})$, the frozen graph has edges

$$\begin{aligned} (--) &\longrightarrow \{(--), (+-)\}, & (+-) &\longrightarrow \{(-+), (++)\}, \\ (-+) &\longrightarrow \{(--)\}, & (++) &\longrightarrow \{(-+)\}. \end{aligned}$$

All coordinates below belong to $X_- \cup X_+$, and their sign-state cycles use these allowed edges. Hence the exact periodic points lie in the source-locked local survivor, not merely in the full planar Hénon map.

Lemma 2.1 (Exact orbit and monodromy ledger). *The survivor contains the following primitive periodic orbits.*

1. *The negative fixed point*

$$q_* = -(1 + \sqrt{7})/6$$

has positive unstable modulus

$$L_1 = 1 + \sqrt{7} + \sqrt{7 + 2\sqrt{7}}$$

and minimal polynomial

$$f_1(X) = X^4 - 4X^3 - 22X^2 - 4X + 1. \quad (3)$$

2. *The period-three sequence*

$$(a, b, a), \quad a = -\frac{\sqrt{5}}{6}, \quad b = \frac{1 + \sqrt{5}}{6},$$

has sign-state cycle $(--, +-, -+)$, return trace $-38 - 42\sqrt{5}$, and positive unstable modulus

$$L_3 = 19 + 21\sqrt{5} + \sqrt{2565 + 798\sqrt{5}}.$$

Its minimal polynomial is

$$f_3(X) = X^4 - 76X^3 - 7374X^2 - 76X + 1. \quad (4)$$

3. *The period-four sequence*

$$(-1/\sqrt{6}, -1/\sqrt{6}, 1/\sqrt{6}, 1/\sqrt{6})$$

has return trace 578 and

$$L_4 = 289 + 24\sqrt{145}, \quad f_4(X) = X^2 - 578X + 1. \quad (5)$$

Proof. Substitution into (2) is direct. The period-three sequence is nonconstant, hence primitive because three is prime. The period-four sequence has no period one or two. Chronological multiplication gives

$$J(a)J(b)J(a) = \begin{pmatrix} -40 - 44\sqrt{5} & 21 + 4\sqrt{5} \\ -21 - 4\sqrt{5} & 2 + 2\sqrt{5} \end{pmatrix},$$

whose determinant is one and trace is $-38 - 42\sqrt{5}$. If $T_3 = 38 + 42\sqrt{5}$, then $L_3 + L_3^{-1} = T_3$. Multiplying the two conjugate quadratics over $\mathbb{Q}(\sqrt{5})$ gives (4). The fixed and period-four calculations give (3) and (5) in the same way. $\square$

3 The degree-32 compositum

Put $K_j = \mathbb{Q}(L_j)$. The key point is not merely that the three multipliers are different; their fields permit independent inversion.

Proposition 3.1 (Field independence). *The compositum*

$$E = K_1 K_3 K_4$$

has degree

$$[E : \mathbb{Q}] = 4 \cdot 4 \cdot 2 = 32.$$

Proof. For K_1 , the quadratic discriminant over $\mathbb{Q}(\sqrt{7})$ is $28 + 8\sqrt{7}$, whose norm is 336, not a square. Moreover $\sqrt{7} = (L_1 + L_1^{-1} - 2)/2$, so $[K_1 : \mathbb{Q}] = 4$.

For K_3 , set

$$d_3 = T_3^2 - 4 = 10260 + 3192\sqrt{5}.$$

Its norm is

$$N_{\mathbb{Q}(\sqrt{5})/\mathbb{Q}}(d_3) = 54323280 = 2^4 3^2 5 \cdot 11 \cdot 19^3. \quad (6)$$

Thus d_3 is not a square in $\mathbb{Q}(\sqrt{5})$, and $K_3 = \mathbb{Q}(\sqrt{5}, \sqrt{d_3})$ has degree four. Finally $K_4 = \mathbb{Q}(\sqrt{145})$ has degree two.

It remains to show that these degrees multiply. The polynomial discriminants are

$$\begin{aligned} \text{disc}(f_1) &= 2^{12} 3^7, \\ \text{disc}(f_3) &= 2^{12} 3^6 5^3 7^4 11 \cdot 19^3. \end{aligned}$$

The field K_1 is therefore unramified at 5, so it cannot contain the ramified quadratic field $\mathbb{Q}(\sqrt{5})$. Hence $[K_1 \mathbb{Q}(\sqrt{5}) : \mathbb{Q}] = 8$.

Because 11 splits in $\mathbb{Q}(\sqrt{5})$ and occurs to odd exponent in (6), one prime above 11 has odd valuation in d_3 . The extension obtained by base-changing K_1 is unramified there, since $11 \nmid \text{disc}(f_1)$. The odd valuation therefore remains odd, so d_3 is still nonsquare after adjoining K_1 . Adjoining $\sqrt{d_3}$ doubles the degree to 16.

Both K_1 and K_3 are unramified at 29, and a compositum of unramified local extensions remains unramified. But 29 ramifies in $\mathbb{Q}(\sqrt{145})$. Consequently K_4 is not contained in $K_1 K_3$, and the final degree is 32. $\square$

The certificate independently computes a primitive element $L_1 + L_3 + L_4$ and an irreducible degree-32 polynomial. This exact machine check supports, but does not replace, Proposition 3.1.

4 Multiplicative and logarithmic independence

Each K_j has a $\mathbb{Q}$ -automorphism ι_j taking L_j to L_j^{-1} . For $j = 1, 3$ it is the sign change of the quadratic square root over the trace field; for $j = 4$ it is conjugation of $\sqrt{145}$. Proposition 3.1 says the three fields are linearly disjoint, so each ι_j extends to E while fixing the other two fields.

Theorem 4.1 (Logarithmic independence). *The positive real numbers*

$$\log L_1, \quad \log L_3, \quad \log L_4$$

are linearly independent over $\mathbb{Q}$.

Proof. After clearing denominators, a rational relation would give

$$L_1^{m_1} L_3^{m_3} L_4^{m_4} = 1$$

with integers m_j . Apply the extension of ι_1 and divide the two equalities. This gives $L_1^{2m_1} = 1$, hence $m_1 = 0$ because $L_1 > 1$. The other two inversion automorphisms similarly give $m_3 = m_4 = 0$. $\square$

5 The all-prime obstruction

We use the real Six Exponentials Theorem: if x_1, x_2, x_3 are linearly independent over $\mathbb{Q}$ and ξ_1, ξ_2 are linearly independent over $\mathbb{Q}$, at least one of the six numbers $\exp(x_i \xi_j)$ is transcendental [1, Theorem 3.1]. This is a theorem, not the open Four Exponentials Conjecture.

Theorem 5.1 (No common-power all-prime triple). *For every real $h > 0$, the three numbers*

$$L_1^h, \quad L_3^h, \quad L_4^h$$

cannot all be rational primes. In fact, if h is rational, none is a rational integer greater than one; if h is irrational, at least one is transcendental.

Proof. First let $h = a/b > 0$ in lowest terms. If $L_j^h = N$ were a rational integer greater than one, then

$$L_j^a = N^b.$$

The left side is an algebraic unit, while a positive rational integer is a unit only when it is one, a contradiction.

Now let h be irrational. Apply the Six Exponentials Theorem with

$$(x_1, x_2, x_3) = (\log L_1, \log L_3, \log L_4), \quad (\xi_1, \xi_2) = (1, h).$$

Theorem 4.1 supplies the first independence hypothesis, and irrationality supplies the second. The three exponentials with $\xi_1 = 1$ are the algebraic numbers L_1, L_3, L_4 . Therefore at least one of the remaining three exponentials L_j^h is transcendental. $\square$

Corollary 5.2 (Pressure labels). *For the actual Hénon pressure root h_* , the labels of the three orbits in Lemma 2.1 cannot all be rational primes. Hence no multiplicity-preserving assignment that declares every intrinsic pressure label $|\Lambda_\gamma|^{h_*}$ to be a rational prime can exist.*

If h_* were algebraic irrational, the stronger Gelfond–Schneider theorem would make each $L_j^{h_*}$ transcendental [1, Theorem 3.2]; the original solution of Hilbert’s seventh problem is due to Gelfond [2]. Corollary 5.2 needs no such hypothesis and also covers transcendental h_* .

6 Exact certificate and adversarial controls

The implementation uses exact radicals, matrices and integer polynomials. It verifies all recurrence equations, determinant-one monodromy, the three minimal polynomials, the discriminant factorizations, and the degree ladder 4, 8, 16, 32. The full degree-32 primitive-element polynomial is stored in the JSON certificate. Its coefficient-vector SHA256 is

d190eee1cc3f950d4d41efdadcc0acedd9c628456d5a1dc6eba8708063c4188a0.

A duplicate-orbit mutation drops the compositum degree from 32 to 16, and a sign mutation of the period-three orbit fails either the trace or the recurrence. Eight inherited artifacts from the pressure, unit, scalar-label and orbit-catalog projects are hash locked.

7 Evaluator verdict and claim boundary

For the termwise pressure-prime lane, the strict Route-A tuple is

$$(A1_{\text{WEAK}}, A2_{\text{FAIL}}, A3_{\text{PARTIAL}}, A4_{\text{FORMAL}}),$$

with overall `ROUTE_A_REJECTED`. Coordinatewise positive evidence from the distinct HCS-P45 dynamical counting object is not merged into this certificate. Route B is not authorized:

no Hilbert space, operator domain, self-adjointness theorem, von Mangoldt trace identity or completed- ξ determinant is supplied.

The obstruction is precisely scoped. It proves that at least one of three labels is not a rational prime; for transcendental h it need not identify which one. It does not refute the pressure prime-orbit law, prime-ideal or Galois-packet labels, cyclic resultants, or a collective signed trace.

8 Conclusion

The pressure-power scalar road is now closed at its strongest termwise interpretation. Three source-native periodic orbits suffice: exact field independence plus a classical transcendence theorem handles every possible real common exponent. The surviving large road is necessarily nonlocal. Arithmetic information must be carried collectively by packets, ideals, resultants or a distributional determinant rather than by one rational prime attached to each primitive orbit.

References

- [1] J. Pila, Geometric and arithmetic postulation of the exponential function, *J. Austral. Math. Soc. Ser. A* **54** (1993), 111–127, [doi:10.1017/S1446788700037022](https://doi.org/10.1017/S1446788700037022).
- [2] A. Gelfond, Sur le septième problème de Hilbert, *Bull. Acad. Sci. URSS, Classe Sci. Math. Nat.* (1934), no. 4, 623–634, <https://www.mathnet.ru/eng/im4924>.
- [3] W. Parry and M. Pollicott, An analogue of the prime number theorem for closed orbits of Axiom A flows, *Ann. of Math.* **118** (1983), 573–591.
- [4] J. Neukirch, *Algebraic Number Theory*, Springer, 1999.